KALAM (கககக) — AGRICULTURAL INSIGHTS ATLAS

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CROPPING INTENSITY

42.2% Double-Cropped Cropland

AUDIT READINESS

205 Targeted Audit Parcels (GPS)

ANNUAL WATER DEMAND

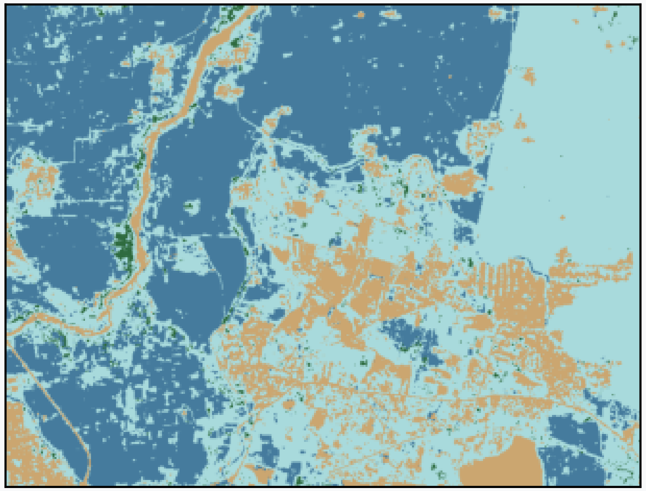
42.64 Million m³ (MCM) Canal Requirement

EXECUTIVE INSIGHT SUMMARY & POLICY ACTION ITEMS

- **Cropping Intensity:** 42.2% of cropland (1269.4 ha) is double-cropped with Kar and Pishanam paddy cycles, with 1.3% under perennial banana.
- **Transplanting Calendar:** Peak Kar transplanting occurs in May (79.3% of Kar fields), while main Pishanam sowing peaks in Nov (70.6% of Pishanam fields). 0.0% of Pishanam acreage experiences delayed transplanting beyond November.
- **Field Verification Priority:** 25.0% of parcels (511 fields) are flagged for targeted ground audit due to lower phenological confidence or inter-annual crop instability, with top-50 priority targets packaged for GPS navigation.
- **Village-Level Statistics:** Kallidaikurichi leads overall agricultural cultivation across both crops with 614.4 ha of paddy (68% share) and 7.7 ha of banana.
- **Irrigation Dependency & Water Access:** 14.7% of paddy acreage is concentrated within 500 m of a river channel, canal or irrigation tank. Banana accounts for 1.4% of cropland in the immediate 0–250 m corridor, decaying sharply beyond 1 km.
- **Crop Change & Farmland Dynamics:** Evaluated 3,011.2 ha of 3,011.2 ha cropland (confidence ≥ 0.60): 52.9% of paddy and 1.3% of banana parcels remained stable across 2023–2026, with 100.5 ha (3.3%) identified as candidate paddy-to-banana diversification.
- **Crop Health & NDVI Anomaly:** 55.0% of analyzed cropland displays normal or above-normal vegetative vigour. Vegetative moisture/vigour stress is observed across 45.1% (1165.6 ha) of cultivated area.
- **Indicative Gross Irrigation Requirement:** Total indicative gross irrigation requirement across analyzed farmed plains is 42.6 MCM/year. Paddy cycles (Kar & Pishanam) account for 88.5% (37.7 MCM), while perennial banana requires 0.8 MCM (1.9%).
- **Segment Size Distribution (Image-Derived):** 19.8% of image-derived field segments (9.9% of agricultural area) are below 1.0 ha. The mean segment size is 1.64 ha with a median of 1.41 ha.

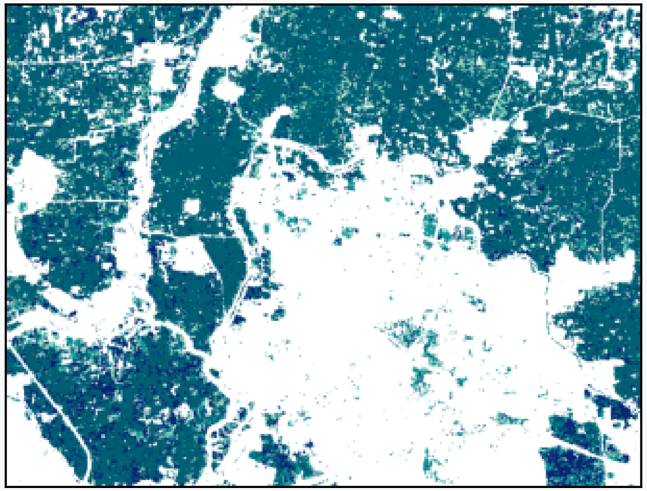
Kalam © 2026 · Problem Statement 1 · SustainAI / Mapathon · Contains modified Copernicus Sentinel data

PHENOLOGY DYNAMICS & VERIFICATION PRIORITIZATION



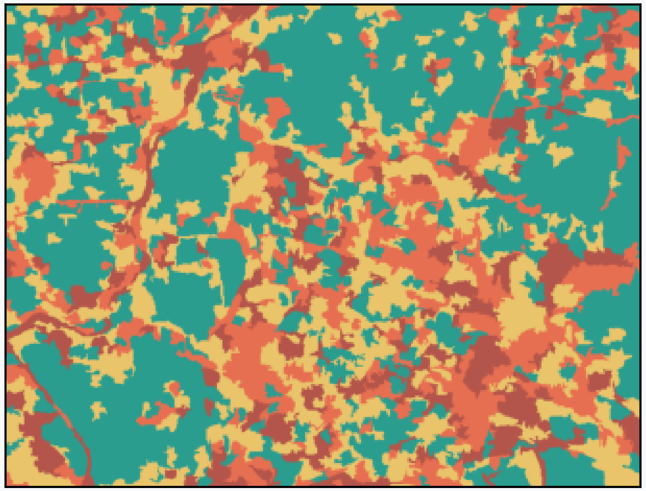
01. Cropping Intensity

tivated (144.8 ha, 4.8%) | Single Crop (1556.5 ha, 51.7%) | Double Crop (1269.4 ha, 42.2%) | Triple Crop (0.0 ha, 0.0%)



02. Transplanting Calendar

Kar Peak (Jun-Jul) | Kar Late (Aug) | Pishanam Peak (Oct) | Pishanam Late (Nov-Dec)

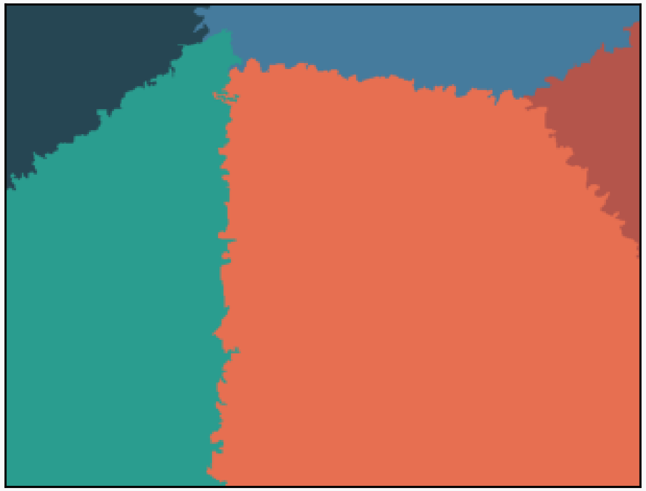


03. Field Audit Priority

Critical Priority (Top 10%) | High Priority (Next 15%) | Medium Priority (Next 25%) | Low Priority (Baseline 50

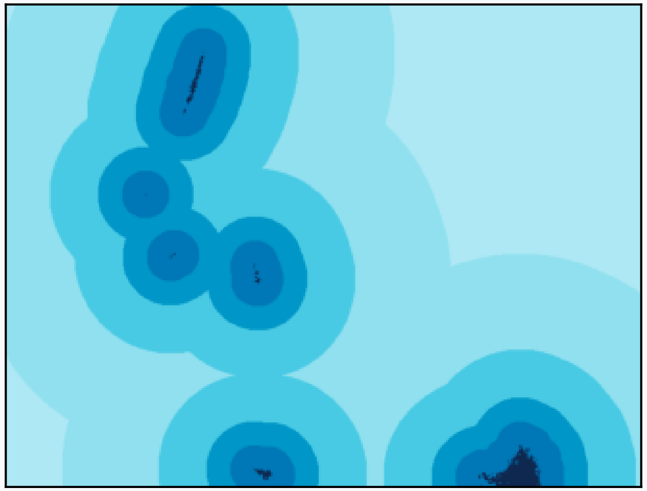
CANAL BUFFER DEPENDENCY & MULTI-YEAR CROP TRANSITIONS

04. Village Crop Allocations



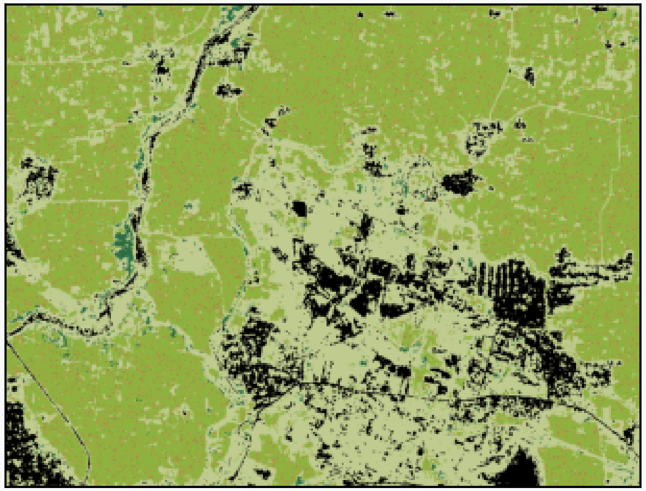
Urban (Paddy intensive) | Kallidaikurichi (Paddy intensive) | Veeravanallur (Diversified / Fallow) | Mukkudal (Paddy intensive)

05. Canal Buffer Decay



0 – 250 m | 250 – 500 m | 500 – 1,000 m | 1,000 – 2,000 m

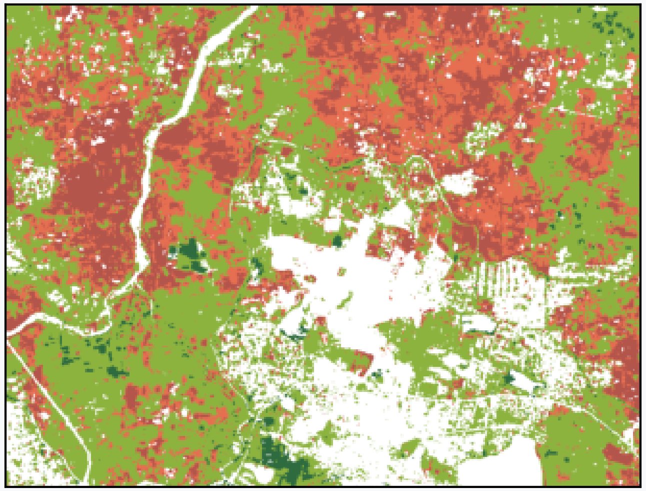
06. Crop Transitions (23–26)



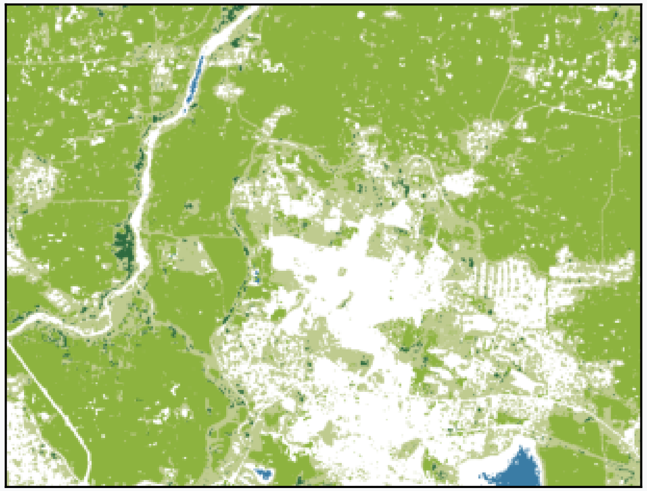
Stable Paddy | Stable Banana | Candidate Paddy → Banana | Banana → Paddy

CROP VIGOUR, WATER DEMAND & PARCEL MORPHOLOGY

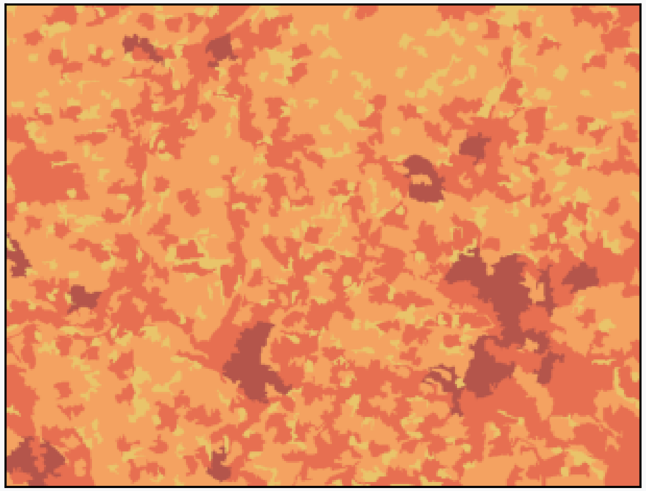
07. NDVI Vigour Anomaly



08. Water Demand Budget



09. Smallholder Fragmentation



ss (530.1 ha, 20.5%) | Stressed [illegible] (50.6% area, 58.1% count) | Me